Initial Project Planning Template

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date
Sprint-1	**Epic 1: Define Problem & Understanding**	**USN-1**	As a researcher, I can identify and define the agricultural problem that the crop recommendation system aims to solve.	2	High	P Hemanth	01 Jun 2026	03 Jun 2026
		USN-2	As a researcher, I can gather and analyze business requirements to understand project objectives and expected outcomes.	2	High	P Hemanth	01 Jun 2026	03 Jun 2026
		USN-3	As a researcher, I can conduct a literature survey to study existing crop recommendation techniques and ML approaches.	2	Medium	P Hemanth	01 Jun 2026	03 Jun 2026
		USN-4	As a researcher, I can analyze the social and business impact of implementing an AI-driven crop recommendation solution.	2	Medium	P Hemanth	01 Jun 2026	03 Jun 2026
	Epic 2: Data Collection & Analysis	**USN-5**	As a developer, I can download the Crop_recommendation.csv dataset from Kaggle and import required Python libraries.	2	High	P Hemanth	04 Jun 2026	07 Jun 2026
		USN-6	As a developer, I can read the dataset and perform univariate, bivariate, and multivariate analysis to understand agricultural data patterns.	3	High	P Hemanth	04 Jun 2026	07 Jun 2026
Sprint-2	**Epic 3: Data Pre-processing**	**USN-7**	As a developer, I can check for null values, detect and handle outliers using IQR method, and extract seasonal crop information.	3	High	P Hemanth	08 Jun 2026	12 Jun 2026
		USN-8	As a developer, I can split the dataset into 80% training and 20% testing sets for model development and evaluation.	2	High	P Hemanth	08 Jun 2026	12 Jun 2026
	Epic 4: Model Building	**USN-9**	As a developer, I can apply K-Means Clustering using the Elbow method to identify patterns and group similar agricultural conditions.	3	High	P Hemanth	13 Jun 2026	18 Jun 2026
		USN-10	As a developer, I can train a Logistic Regression model, evaluate its performance using classification metrics, and save the best model as model.pkl.	3	High	P Hemanth	13 Jun 2026	18 Jun 2026
		USN-11	As a farmer, I can predict the most suitable crop by providing N, P, K, temperature, humidity, pH, and rainfall values to the trained model.	2	High	P Hemanth	13 Jun 2026	18 Jun 2026

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Sprint-3	**Epic 5: Application Building**	**USN-12**	As a developer, I can design and build HTML pages (home.html, about.html, findyourcrop.html) for the OptiCrop web application.	3	High	P Hemanth	19 Jun 2026	24 Jun 2026
		USN-13	As a developer, I can build the Python Flask backend (app.py) and integrate it with the trained ML model for real-time crop prediction.	3	High	P Hemanth	19 Jun 2026	24 Jun 2026
		USN-14	As a farmer, I can run and test the OptiCrop web application, enter soil parameters, and receive accurate crop recommendations instantly.	2	High	P Hemanth	25 Jun 2026	26 Jun 2026